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RESEARCH ARTICLE

Dietary exposure to polyvinyl chloride microparticles induced oxidative stress and hepatic damage in *Clarias gariepinus* (Burchell, 1822)

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Abstract

The present study investigated the effects of polyvinyl chloride (PVC) microparticles (MP) on hepatic antioxidant enzymes activities, serum biochemical and liver histology of juvenile *Clarias gariepinus*. A total of 180 (25.15 g average weight) *C. gariepinus* were fed PVC MP (95.41 ± 4.23 μm) spiked diets at 0.5, 1.5, 3.0 percentage inclusion levels and a control diet for 45 days of exposure, then followed by 30 days of depuration trials. Fish specimens (9) from each treatment were sampled every 15-day interval for serum biochemical, liver antioxidant enzymes and histopathological assay. Glucose and triglyceride levels increased significantly in PVC-treated groups when compared with the control. Protein levels of 0.5% and 3.0% PVC-treated groups reduced significantly on the 15th and 30th day exposure periods, while serum enzyme activities of all PVC-treated groups increased significantly in a time-dependent manner. Antioxidant enzymes (superoxide dismutase, glutathione peroxidase, catalase) activity in the liver of the treated groups also decreased progressively in a time-dependent manner. A time-dependent elevation in lipid peroxidation levels was observed in PVC MP-treated groups. Histopathological assessment of the fish liver showed mild to severe levels of glycogen depletion, fatty vacuolation and degeneration, hepatocellular necrosis in PVC-treated groups with reference to the control. The present study revealed that PVC microplastic induced oxidative damage and hepatic histopathological alterations in the exposed fish.

Keywords Ecotoxicology · Microplastics · Biomarkers · Histopathology · African catfish

Introduction

Plastic materials exist in different forms and are used daily by individual for diverse purpose (Pellini et al. 2018; Arias-Andres et al. 2018). Due to its low cost, durability, availability and flexibility (World Economic Forum 2016; Pellini et al. 2018; Jovanovic et al. 2018), its demand in different forms have increased geometrically. Wright et al. (2013) stated that 1.5 million to 322 million tons were produced between 1950 and 2015. Global production of petroleum-based plastic exceeded 300 million tons in 2015 (Wu et al. 2017). The most challenging and disturbing aspect of plastic lies on its pollution potentials within the environment (Abayomi et al. 2017). According to Jovanović (2017), plastic-based environmental pollution will be the biggest challenge scientists and risk managers of the next generation might likely face. Plastics enter into the aquatic environment via two major means, namely the primary and secondary sources (Galloway et al. 2017; Gallo et al. 2018). The primary sources include the discharge of industrial effluent into streams and rivers, waste from medical facilities and house hold microbeads-based waste products. Additionally, abandonment of fish gears during fishing and accidental spillage of plastic resin pellets and materials during aquatic transport have also been identified as primary sources (Ogata et al. 2009; Andrady 2011). The secondary sources involve the breakdown of macroplastics (> 5 mm) into microplastics and nanoplastics via different chemical processes such as fragmentation, photo-degradation and the weathering processes (Kettner et al. 2017; Pellini et al.

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2018), thereby intensifying the concentration of microplastic (< 5 mm) in both freshwater and marine ecosystems (Jovanović 2017; Ribeiro et al. 2017). Microplastics have been reported to be the most prevalent type of contaminant found in oceans, fresh water lakes, beaches and tributaries (NOAA 2016; Horton et al. 2017).

Polyvinyl chloride (PVC) with the chemical formula (C_2 H_3 Cl) n is a thermoplastic made of 57% chlorine and 43% carbon and is the second most used synthetic plastic polymer for the production of different forms of plastics worldwide after polyethylene, and this is due to its long-term durability, weather ability and electrical properties (Terry 2012; Tiwari 2015). PVC is used in the production of toys, jars, detergent bottles, medical tubes and other house hold products (Terry 2012; Mahon et al. 2017). In recent times, PVC materials in forms and sizes have been traced and documented to be found in water sediments, column and surface of different water bodies of both fresh and marine origin (Tiwari 2015; Jovanović 2017; Foley et al. 2018).

The presence of plastics in the aquatic environment poses serious health risk issues to both flora, fauna and humans at large (Santillo et al. 2017; Zhang et al. 2017; Scherer et al. 2018). Plastics are ingested by aquatic organisms accidentally or intentionally (Jovanović 2017; Abbasi et al. 2018), due to their appealing coloration, buoyancy and semblance to food (Romeo et al. 2015). Uptake of plastics of different size ranges by fish species has been reported (Lusher et al. 2015; Nadal et al. 2016; Güven et al. 2017). An in vitro experiment conducted by Ribeiro et al. (2017) revealed the accumulation of microplastics in the gill and digestive gland of clams (*Scrobicularia plana*) after 14-day exposure trial. Additionally, the presence/accumulation of microplastics was evident in the liver of gilt-head sea beam (*Sparus aurata*) fed microplastic spiked diet (Jovanovic et al. 2018) after 45-day exposure period. The aftermath effect of plastic uptake by aquatic organisms are X-rayed or manifested behavioural and physiological levels (Cole et al. 2015; Avio et al. 2015) such as blockage of the digestive organs (Jovanović 2017), injuries on the intestinal wall, ulceration, shortening and swelling of villi, loss of regular structure of serosa and hyperplasia of globlet cells (Pedà et al. 2016). Others include increase in degranulation of primary granules (Greven et al. 2016), alteration of lipid metabolism in fish (Cedervall et al. 2012), fatty vacuolation, glycogen depletion and cell necrosis in the liver of exposed fish (Rochman et al. 2013). Oliveira et al. (2013) reported significant reduction in acetylcholinesterase activity, indicating neurotoxicity in fish. Oxidative stress was reported as a result of increase in catalase and superoxide dismutase activity in plastic exposed fish (Li et al. 2015).

Clarias gariepinus is an important commercial species and highly pleasured in Nigeria and other African countries. *C. gariepinus* is an omnivorous species and feast on any food material aside plants and fish. The fish species inhabits almost all inland fresh waters including flooding plains and swamps (Iheanacho et al. 2018). *C. gariepinus* is hardy and can withstand tough environmental situations, hence is often used as model fish for toxicological studies (Idodo-Umeh 2003; Nwani et al. 2016). Okoro et al. (2019) studied the toxicity of *Chromolaena odorata* leaf extract (Siam weed found alongside the rivers and streams) on aquatic organisms using *C. gariepinus* as bioindicator. The effects of ibuprofen (Ogueji et al. 2018), propanil herbicide (Yaj et al. 2018), carbendazim fungicide (Ezeoyili et al. 2019), cyperdicot insecticide (Odo et al. 2017), plastic microparticles (Iheanacho and Odo 2020) and polycyclic aromatic hydrocarbons (Sogbanmu et al. 2018) on the health of aquatic biota have been assessed using *C. gariepinus* as bio model. The present study investigated the effect of PVC microparticles (being one of leading plastic used worldwide) on aquatic species, using *C. gariepinus* as bioindicator and assessed via a battery of biomarkers of oxidative stress (catalase, superoxide dismutase, glutathione peroxidase), serum enzymes (alkaline phosphatase, aspartate aminotransferase, alanine aminotransferase), oxidative damage (lipid peroxidation) and histopathology.

Materials and methods

Biological model and rearing condition

Healthy African catfish juveniles *C. gariepinus* ($N = 180$) were procured from the Department of Fisheries and Aquaculture farm, Alex Ekwueme Federal University Ndufu Alike, Nigeria. Fish specimens were immediately transferred to the departmental wet laboratory. On arrival at the laboratory, the fish were first quarantined in a holding fish tank prior to the acclimation period. Fish were acclimated in experimental tanks for 2 weeks prior to the beginning of the experiment and were fed with commercial fish diet (Coppens International, Netherlands) at 3% body biomass two times (8:00–18:00 h) daily. The culture media was renewed every 3 days in order to maintain optimal water quality conditions. The physicochemical parameters of the culture media were estimated daily, using water quality kit (ProLab™, Florida), and values were maintained at dissolved oxygen 5.20 ± 0.23 mg/L; temperature 27.47 ± 0.35 °C; pH 7.12 ± 1.34; conductivity 42.71 ± 1.56 Us/cm and ammonia 0.04 ± 0.12 mg/L, with a controlled photoperiod of (12 h light:12 h darkness) in the laboratory.

Polyvinyl chloride particle size determination

Purity grade of polyvinyl chloride (99.5% purity, powdered form), under the trade name Formolon® 622 (CAS no. TH1764490248), was sourced from a plastic manufacturing firm in Enugu state, Nigeria. Particle size was achieved using optical microscopy (Nikon Eclipse E100, China) (Plate 1) and

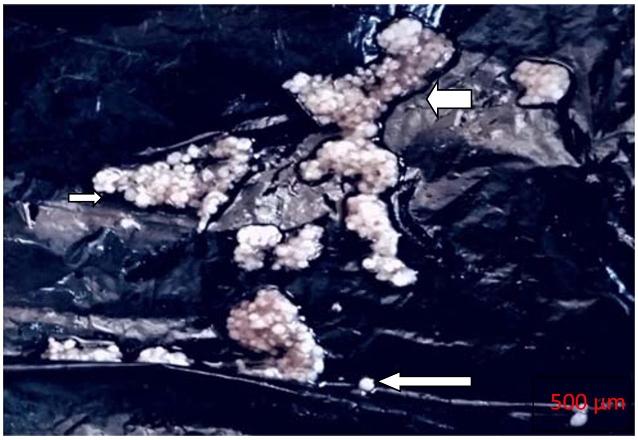

Plate 1 Polyvinyl chloride microplastics (white arrow)

an automated laser particle size analyser (Microtrac S3500, 0.02–2800 µm, Germany). Small sample (200 mg/g^{-}kg^{-1}) of PVC powder was fed into the dry dispersion device (Turbotrac) to deliver dispersed samples to the measuring cell in the Microtrac optical bench for particle size analysis. The statistical data regarding the percentage of particle size in various ranges was ascertained from the instrument's software.

Ethical statement

All experimental procedures were performed in compliance with the standards described by the institution of animal welfare act in line with the National Environmental Standard Regulations Enforcement Agency (NESREA) Act of Nigeria on the protection of animals against cruelty.

Diet preparation

Four iso-caloric and iso-nitrogenous diets were formulated for the experiment. The different feed stuffs and their inclusion levels are as follows: fish meal 40%, corn meal 15%, soybean meal 15%, fish oil 1.0%, wheat offal 15%, vitamin/mineral premixes 9% and starch (binder) 5%. PVC MP were incorporated into the diets as treatments, with the following inclusion percentages of 0.0% (control), 0.5%, 1.5% and 3.0%. Though the amount of plastic pellets/resins ingested by fish in the wild is not known (Graca et al. 2017; Jovanović 2017), percentage dietary incorporations of PVC resins in the present study were within the concentration ranges (0.5–10% of diet) reported in the previous studies (Rochman et al. 2013; Santana et al. 2016; Jovanovic et al. 2018). Formulated fish diets were mechanically mixed and pelletised into 3-mm diameter size using a locally fabricated machine. Pelleted diets were air dried at 25 °C for 24 h and stored at 4 °C in the refrigerator until used. Approximate compositions of the fish diets are as follows: crude protein 50.10%, crude fat 15.90%, crude fibre 5.20%, crude ash 10.25%, moisture 5.0% and nitrogen free extract 13.55%.

Experimental protocol and microparticle exposure

The method of PVC exposure to fish was reproduced as published by Jovanovic et al. (2018) and OECD (2012)). The dietary method became imperative due to the indissolubility of PVC resin in water and organic solvents such as acetone and dimethyl sulfoxide (Santana et al. 2016; Jovanovic et al. 2018). Sedimentation propensities of PVC, polystyrene, polythene in the aquatic environment and their bioaccumulation in aquatic organisms have also been reported in previous studies (Oliveira et al. 2013; Lu et al. 2016). A total of 180 juveniles *C. gariepinus* (25.15 ± 1.34 g) were randomly assigned to four test diets containing 0% (control), 0.5%, 1.5% and 3.0% PVC MP, in a completely randomised design (CRD) pattern. Each treatment contained 45 fishes and was further randomised into three replicates of 15 fishes per replicate. Feeding of fish was done twice (morning and evening) daily at 3% body weight. The feeding trial lasted for 45 days known as the exposure period/phase (fish were exposed to PVC spiked diet), followed by 30-day depuration period (fishes were fed to the control diet).

Organosomatic index measurement

Fish specimens (3) from each experimental tank were sampled biweekly to ascertain the weight of fish. The fish specimens were first sampled for blood collection followed by liver tissue sampling. The liver tissue was weighed using sensitive weighing balance (S. melter, 1–100 mg, China). Values obtained were used to estimate the hepatosomatic index according to Sogbanmu et al. (2018);

$$\text{Hepatosomatic Index (HSI)} = \frac{\text{fish liver weight (g)}}{\text{fish body weight (g)}} \times 100$$

$$\text{Survival Rate (\%)} = \frac{\text{Number of fish that survived}}{\text{Total number of fish stocked}} \times 100$$

Procedure for blood collection

Three fishes per replicate ($n = 9$ per treatment) were sampled at 15 days of interval for blood collection during the exposure trial (45 days), while at the end of the depuration trial (30 days), the same number of fish was also sampled for blood collection. The fish were randomly caught using hand net and anaesthetized with tricaine methane sulfonate MS 222 (CAS no: 886-862, Tricaine S®, Aqualife Sciences Inc., USA) at 50.0 mg/L to minimise stress. The method for blood collection was done following the procedure of Bello et al. (2014). Sterilised 3G needles and syringes of 5 ml were used to collect blood from fish. At first, the ventral cavity of the fish was wiped with dry tissue paper to avoid contamination with mucus during blood collection. At a distance of 3–4 cm from the genital opening of each fish, the needle was inserted at a right angle to the vertebral column of the fish. Under gentle aspiration, 3 ml of blood was taken and needle was gently withdrawn and transferred into plain tube for serum chemistry analysis.

Serum biochemical analysis

The clotted blood in the plain tubes was transferred into clean dry centrifuged tubes with Hawsley minor bench centrifuge at 3000 r/min for 15 min (Hesser 1960). The supernatant (serum) formed after centrifugation was collected using Pasteur pipette and transferred into a plain plastic test-tube for biochemical analysis. Biochemical parameters such as total protein, glucose level, triglycerides and serum enzymes (alkaline phosphatise (ALP), alanine aminotransferase (ALT) and aspartate aminotransferase (AST) were determined using a semi-automated biochemistry analyser (Sinothinker SK3002B, China). Reagents and procedures for the analysis were strictly adhered to, according to the manufacturer's instructions.

Tissue collection and preparation

After the blood collection, the same fish were further sacrificed for tissue collection. Liver tissue was carefully collected using spatula to avoid damage of tissues after evisceration, hence rinsed in physiological saline to remove traces of blood. Tissue samples were transferred to the laboratory for antioxidant enzymes assay and histopathological examination.

Antioxidant enzymes assay

Superoxide dismutase (SOD) activity in fish was estimated according to the method of (Misra and Friedovich 1972). A 0.1-ml tissue homogenate was added to the tubes containing 0.75 ml of ethanol and 0.15 ml of chloroform under chilled condition and centrifuged. To 0.5 ml of supernatant, 0.5 ml 0f 0.6 Nm EDTA solution and 1.0 ml of 0.1 M carbonate-bicarbonate buffer (pH 10.2) were added. The reaction was initiated by the addition of 0.5 ml 0f 1.8 mM epinephrine and the increase in absorbance at 30-s interval for 3 min was measured at 480 nM in a Shimadzu UV spectrophotometer. One unit of superoxide dismutase activity was expressed as the amount of protein required for 50% of inhabitation of epinephrine autoxidation/min.

Glutathione peroxidase (GPx) activity in fish liver was measured according to the method of Beutler (1984). Three grammes of tissue was homogenised in 2-ml Tris buffer. The reaction system contained 0.2-ml Tris buffer, 0.2-ml EDTA, 0.1-ml sodium azide and 0.5 ml of tissue aliquot. To this, mixture was added 0.2 ml of glutathione and 0.1 ml of hydrogen peroxide and shaken very well. The reaction mixture was incubated for 10 min at 30 °C. A control blank containing all the reagents except the test tissue homogenate was also set up. After 10 min, 0.5 ml of 10% TCA was added to and mixed before centrifuging the mixture. The supernatant was used to assay for GPx by reading the absorbance at 412 nm and the values was given as µg/min/mg protein.

Catalase activity was estimated according to the method of Beutler (1984). To 1.2 ml of 0.01 mM phosphate buffer (pH 7.0), 0.5 ml tissue homogenate was added. The enzyme reaction started by the addition of 1.0 ml of 0.2 mM hydrogen peroxide (H_2O_2) solution. The decrease in absorbance was measured at 250 nm for every 30 s up to 3 min. The enzyme blank was run simultaneously with 1.0 ml of distilled water instead of hydrogen peroxide. The enzyme activity was expressed as µmole of hydrogen peroxide decomposed/min/mg of protein.

Lipid peroxidation estimation

The estimation of thiobarbaturieacide reactive substance (TBARS) evaluated as malondialdehyde content of the liver tissue was measured according to the method of Sharma and Krishna-Murti (1968). A 3-g tissue was homogenised in Tris-HCI buffer and mixed with 2-ml TCA-TBA-HCI reagent. One millilitre of distilled water was added to the reaction mixture and heated in a water bath for 15 min. The reaction system was cooled and centrifuged at 600*g* for 10 min. The colour of the supernatant was read at 525 nm against a reagent blank. The extinction coefficient of malondialdehyde is given as 1.56×10^5 M^{-1} cm^{-1}. The value obtained was expressed as mmol/100 g tissue.

Histopathology

At the end of the study, the appropriate samples were collected for histopathological examination. The samples were fixed in 10% phosphate buffered formalin for a minimum of 48 h before commencement of tissue preparation processess. The tissues were trimmed, dehydrated in 4 grades of alcohol (70%, 80%, 90% and absolute alcohol), cleared in 3 grades of xylene and embedded in molten wax. On solidifying, the blocks were sectioned, 5 µm thick with a rotary microtome, floated in water bathe and incubated at 60 °C for 30 min. The 5-µm thick sectioned tissues were subsequently cleared in 3 grades of xylene and rehydrated in 3 grades of alcohol (90%, 80% and 70%). The sections were then stained with haematoxylin for 15 min. Blueing was done with ammonium chloride. Differentiation was done with 1% acid alcohol before counterstaining with Eosin. Permanent mounts were made on degreased glass slides using a mountant; DPX. The prepared slides were examined with a Motic™ compound light microscope using ×4, ×100 and ×400 objective lenses. The photomicrographs were taken using a Motic™ 9.0 megapixels microscope camera at ×100 and × 400 magnifications. Observations made based on cytomorphological changes in the tissues was deduced subjectively (under eye view). A list of histopathological features (Jovanovic et al. 2018) was assessed and the degree of severity designated as none, mild, moderate and severe (Table 5).

Statistical analysis

Data generated from the parameters measured were presented as mean ± standard deviation (SD). All data were preliminary tested for normality (Kolmogorov-Smirnov and Shapiro-Wilk test) and homogeneity (Levene test) prior to analysis. Statistical difference was estimated using one-way analysis of variance (ANOVA) SPSS IBM version 22.0 computer program (SPSS Inc. Chicago, Illinois, USA), followed by post hoc Ducan multiple range test (DMRT). A two-way analysis of variance was also conducted to verify significance between treatments, exposure time and interaction effects. Microsoft excel (version 16.0) was used to present results in bar charts, where necessary.

Result

Microparticle size

Photo of PVC microparticles used in the present study (dietary exposure) is shown in Plate 1. The average size of PVC microparticle used in the present study was 95.41 ± 4.23 μm.

Hepatosomatic index and body biomass measurement

Result on hepatosomatic index, fish body biomass and survival rate of fish specimens treated to various concentrations of PVC spiked diets for 45 days is presented in Fig. 1a–c. Hepatosomatic index values of the PVC-treated groups were significantly higher ($p < 0.05$) than the control (Fig. 1a). After 45-day exposure periods, variations in body weight (Fig. 1b) and survival rate (Fig. 1c) was not significantly different ($p > 0.05$) among the treated groups relative to the control.

Serum biochemistry

Effects of PVC MP on biochemical parameters of *C. gariepinus* juvenile are shown in Tables 1 and 2. In terms of glucose, a significant difference ($p < 0.05$) was observed among the treated groups, likewise between the different exposure durations and interaction between the treatment and duration. Serum glucose levels of the treated groups were significantly higher ($p < 0.05$) than the control. A similar trend was also observed during the post exposure period. Assessment of total protein levels revealed that there were significant differences ($p < 0.05$) between treatments, exposure durations and interaction effect (Table 2). Total protein levels of 0.5% and 3.0% treated groups were significantly lower than the control on the 15th and 30th day exposure periods, but on the 45th day, no significant difference was observed among treated groups with reference to the control. The protein level of 1.5% treated group reduced significantly ($p < 0.05$) and was time dependent. Changes in triglyceride levels of fish were found to be significantly different ($p < 0.05$) among treatments, but not significant ($p > 0.05$) assessing the different exposure durations as well as the interaction effect. The triglyceride level of the control fish differed significantly from the treated groups on the 30th and 45th day exposure periods, but was observed to be insignificantly different on the 15th day exposure time. The depuration trial showed that the triglyceride (TG) levels of the treated groups were significantly higher than the control.

Effect of PVC MP on serum enzymes (ALP, AST and ALT) activity in *C. gariepinus* is presented in Tables 1 and 2. The findings of the present study revealed that ALP activity of the control was not significantly different from the treated groups, but increased significantly in a time-dependent manner. No significant changes were observed in ALP activities among the treated groups when compared with the control, at the end of the depuration period and the interaction effect was also insignificant. Information regarding ALT activity showed that there was significant difference among treatments, exposure durations, likewise the interaction effects. On the 15th day, ALT activity of 0.5% treated group increased significantly than the control, whereas no significant change was found among the treated groups with reference to the control on the 30–45th day exposure time. Assessment of ALT activity with regard to the different exposure periods revealed that its activity increased significantly in all concentrations in a time-dependent manner. The activity ALT was also observed to be significantly higher in treated groups than the control at the end of depuration period. The findings of the present study concerning the activity of AST revealed that there was significant difference among treatments, exposure durations and interaction effect. AST activity increased significantly in the treated groups than the control. A time-dependent increase in the activity of AST was seen in 1.5 and 3.0% treated fish. AST activity of the control was observed to be significantly different from the treated fish at the end of the depuration period.

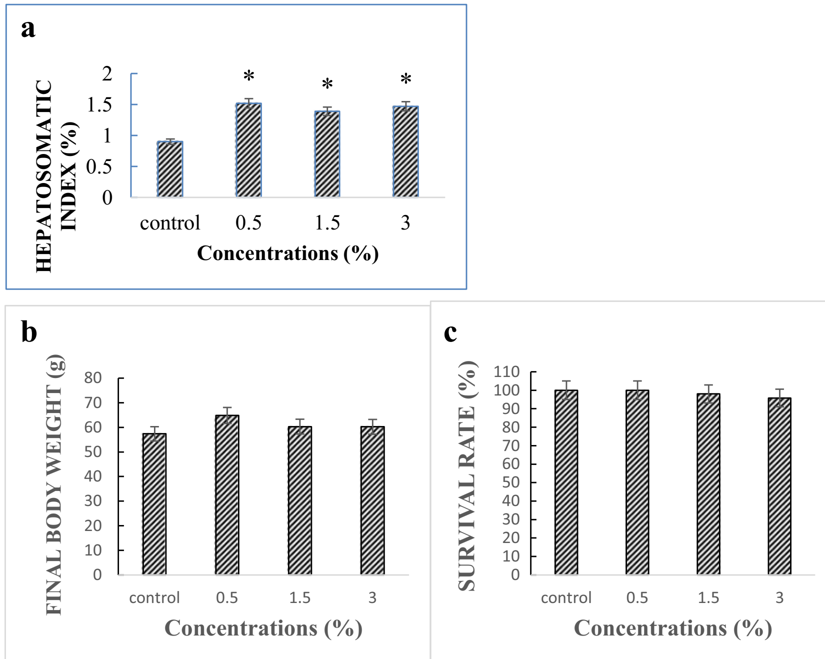

Fig. 1 Data on hepatosomatic index (a), average weight gain (b) and survival (c) of *Clarias gariepinus* juvenile treated to different concentrations of poly vinyl chloride MP spiked diets after 45 days of exposure period. Bars represent means and vertical lines the SE of nine individual observations. Bars with asterisk (*) denote significant difference ($p < 0.05$) (based on one-way ANOVA)

Oxidative stress indices

Response of liver antioxidant enzymes (SOD, GPx, CAT) of *C. gariepinus* treated with PVC MP spiked diets is presented in Tables 3 and 4. The activity of SOD in PVC treated group did not differ significantly ($p > 0.05$) from the control. Assessment on SOD activities between different exposure times showed a biphasic trend, while SOD activity assessed at the end of the depuration period showed no significant difference ($p > 0.05$) between the control and the treated groups, but the interaction effect was significant. Activity of GPx was significantly inhibited in PVC MP-treated groups when compared with the control on the 30th day, but no significant changes ($p > 0.05$) were observed between the control and the treated groups on the 45th day and during the post exposure period. Meanwhile, GPx activity reduced progressively ($p < 0.05$) in all concentration treated groups and was time-dependent. Assessment on the activity of CAT in the liver revealed no significant changes between the control and PVC-treated groups throughout the experimental period, but reduced significantly in 1.5 and 3.0% treated fish specifically on the 45th day exposure.

Lipid peroxidation

A progressive and concentration dependent increases in LPO levels were observed in all PVC-treated groups with reference to the control on the 15th and 45th day, but was not significantly different from the control on the 30th day. However, LPO level increased progressively ($p < 0.05$) in all concentration treated groups and was observed to be time dependent, but the interaction effect was not significant (Tables 3 and 4).

Histopathology

Liver histopathology micrographs of juvenile *C. gariepinus* treated with different percentage inclusion levels of PVC MP spiked diets between the 15th and 45th day exposure periods as well as the depuration period (30 days) are shown in Figs. 2, 3, 4 and 5 and Table 5. On the 15th day exposure (Fig. 2), the photo micrograph sections of the control (A) and all treated groups (0.50%) (B), (1.50%) (C) and (3.0%) (D) showed normal histo-architecture of the piscine liver. The hepatic parenchymas are contained within a thin capsule of fibrous connective tissue. The parenchyma showed normal polyhedral-shaped hepatocytes with central nuclei and abundant glycogen-rich cytoplasm. The hepatocytes are arranged in irregular sheets and/or bundles separated by thin sinusoidal spaces. However, mild infiltration of hepatic cells into the central vein was observed in section 3.0% treated group (D). On the 30th day exposure (Fig. 3), the control (A) and 0.50% treated fish (B) showed normal hepatic histo-architecture of the piscine liver which is glycogen-rich. However, 1.50% treated fish (C) showed mild hepatocytic glycogen depletion while the liver of 3.0% treated fish (D) showed mild to moderate hepatocytic glycogen depletion and fatty vacuolation (arrow). On the 45th day exposure (Fig. 4), liver of the control fish (A) showed normal hepatic histo-architecture of the piscine liver. Moderate hepatocytic glycogen depletion and infiltration of hepatocytes into the central vein was observed in

Table 1 Serum biochemical profile of *Clarias gariepinus* juvenile treated to different concentrations of poly vinyl chloride MP spiked diets

Parameter	Conc. (%)	Exposure period (days)			
		15	30	45	30-day depuration
Glucose (*mg/dl*)	Control	65.51 ± 3.16^{bA}	64.89 ± 3.09^{bA}	65.97 ± 3.49^{bA}	65.82 ± 3.32^{cA}
	0.50	77.99 ± 2.21^{aA}	72.75 ± 6.12^{aB}	72.35 ± 5.05^{aB}	73.75 ± 5.09^{aA}
	1.50	69.12 ± 2.63^{bA}	68.80 ± 1.71$^{ab\ A}$	70.06 ± 2.33^{aA}	69.83 ± 2.32^{bA}
	3.00	75.06 ± 1.95^{aA}	72.67 ± 2.94^{aB}	73.28 ± 2.53^{aA}	73.73 ± 2.45^{aA}
Protein (*g/gl*)	Control	5.04 ± 0.28^{aA}	4.31 ± 0.86^{aA}	3.91 ± 0.93^{aAB}	4.19 ± 0.95^{aA}
	0.50	3.52 ± 0.43^{bA}	3.47 ± 0.28^{bA}	3.30 ± 0.34^{aA}	3.36 ± 0.36^{bA}
	1.50	4.35 ± 0.05^{abA}	3.93 ± 0.48^{abA}	3.63 ± 0.59^{aB}	3.81 ± 0.60$^{ab\ A}$
	3.00	3.85 ± 0.90^{bA}	3.73 ± 0.59^{abA}	3.53 ± 0.56^{aA}	3.61 ± 0.63^{bA}
Tryglycerides (*mg/dl*)	Control	210.77 ± 22.47^{aA}	210.54 ± 16.09^{bA}	211.19 ± 14.11^{cA}	211.14 ± 12.78^{bA}
	0.50	212.59 ± 11.54^{aA}	218.15 ± 14.19^{abA}	220.77 ± 14.57^{bcA}	219.19 ± 12.88^{abA}
	1.50	232.36 ± 15.19^{aA}	230.07 ± 10.39^{aA}	230.09 ± 8.56^{abA}	226.26 ± 12.44^{aA}
	3.00	234.44 ± 7.02^{aA}	232.57 ± 7.38^{aA}	232.74 ± 7.32^{aA}	229.12 ± 9.85^{aA}
ALP (*U/I*)	Control	31.10 ± 9.88^{aAB}	42.98 ± 3.30^{aA}	37.13 ± 9.33^{aA}	34.09 ± 9.03^{aAB}
	0.50	27.94 ± 4.37^{aB}	46.54 ± 0.87^{aA}	37.34 ± 10.67^{aA}	34.34 ± 9.77^{aB}
	1.50	25.87 ± 1.59^{aB}	45.55 ± 4.37^{aA}	35.81 ± 11.26^{aA}	32.76 ± 10.03^{aB}
	3.00	24.88 ± 4.00^{aB}	48.03 ± 1.17^{aA}	36.55 ± 13.05^{aA}	32.92 ± 11.69^{aB}
ALT(*U/L*)	Control	25.33 ± 2.52^{bB}	34.00 ± 9.75^{aA}	38.06 ± 9.92^{aA}	36.63 ± 9.11^{bA}
	0.50	32.33 ± 2.08^{aC}	40.83 ± 9.45^{aB}	44.88 ± 9.66^{aA}	45.74 ± 8.41^{aA}
	1.50	27.67 ± 3.06^{bB}	38.83 ± 12.45^{aA}	43.87 ± 12.44^{aA}	45.90 ± 11.34^{aA}
	3.00	29.00 ± 1.00^{abB}	38.50 ± 10.56^{aA}	42.85 ± 10.69^{aA}	45.22 ± 10.13^{aA}
AST (*U/L*)	Control	29.00 ± 5.57^{bAB}	38.67 ± 11.47^{bA}	43.80 ± 12.10^{bA}	39.85 ± 12.60^{bA}
	0.50	54.67 ± 7.51^{aA}	52.33 ± 5.65^{aA}	53.33 ± 4.99^{aA}	48.94 ± 9.29^{abB}
	1.50	46.67 ± 8.08^{aB}	49.17 ± 5.88^{aA}	52.11 ± 6.46^{aA}	46.67 ± 11.37^{abB}
	3.00	53.00 ± 3.61^{aB}	53.33 ± 3.01^{aB}	55.43 ± 4.29^{aA}	49.82 ± 10.79^{aC}

ALP, alkaline phosphatase; *ALT*, alanine aminotransferase; *AST*, aspartate aminotransferase. Data ($n = 9$) are presented as mean ± SD. Values with different lowercase letters superscript between control and different concentrations differ significantly ($p < 0.05$) within the same exposure period; values with different uppercase letters superscript differ significantly ($p < 0.05$) among different exposure periods within the same concentration

the liver of 0.50% treated fish. The liver of 1.50% PVC treated fish (C) showed moderate hepatocytic glycogen depletion and fatty degeneration (arrow). The liver section of 3.0% treated fish (D) showed severe hepatocytic glycogen depletion. In addition, moderate fatty degeneration, infiltration of hepatic cells into the central vein and single-cell necrosis were also observed in the liver of 3.0% treated fish (D) (arrow). During the 30th day of recovery period (5), the liver of the control fish

Table 2 A two-way ANOVA statistics for serum biochemical parameters of *Clarias gariepinus* exposed to polyvinyl chloride MP spiked diets

Main effects				Interaction			
Parameter		Treatment		Exposure duration		Treatment × duration	
	R squared	*F*	*P* value	*F*	*P* value	*F*	*P* value
Glucose	0.734	29.072	0.000*	8.946	0.000*	3.437	0.005*
Protein	0.693	9.396	0.000*	23.462	0.000*	2.495	0.027*
Triglycerides	0.169	5.019	0.006*	1.733	0.180	0.478	0.878
ALP	0.814	0.324	0.808	78.477	0.000*	0.988	0.455
AST	0.872	14.672	0.000*	83.348	0.000*	4.657	0.001*
ALT	0.935	35.708	0.000*	178.745	0.000*	5.831	0.000*

Values marked with * denotes significant difference at 5% ($p > 0.05$). *ALP*, alkaline phosphatase; *ALT*, alanine aminotransferase; *AST*, aspartate aminotransferase

Table 3 Antioxidant enzyme activities (mmol min^{-1} mg $protein^{-1}$) and lipid peroxidation in the liver of *Clarias gariepinus* juvenile treated to different concentrations of poly vinyl chloride MP spiked diets

Parameter	Exposure period (days)				
	Conc. (%)	15	30	45	30-day depuration
SOD	Control	43.72 ± 6.16^{aA}	44.83 ± 2.26^{aA}	39.98 ± 1.53^{aA}	43.37 ± 6.11^{aA}
	0.50	39.93 ± 4.98^{aAB}	41.32 ± 3.90^{aA}	36.85 ± 3.48^{aB}	39.61 ± 4.95^{aAB}
	1.50	41.07 ± 3.02^{aB}	42.39 ± 2.87^{aA}	37.80 ± 2.56^{aC}	40.74 ± 2.99^{aB}
	3.00	40.63 ± 3.51^{aB}	44.02 ± 3.25^{aA}	39.27 ± 2.89^{aC}	40.31 ± 3.49^{aB}
GPx	Control	29.60 ± 4.15^{aA}	24.67 ± 1.55^{aAB}	21.90 ± 1.55^{aAB}	27.23 ± 3.79^{aA}
	0.50	25.22 ± 5.03^{aA}	21.06 ± 1.03^{bC}	21.81 ± 2.31^{aC}	23.20 ± 4.63^{aB}
	1.50	28.40 ± 5.08^{aA}	22.10 ± 1.32^{bD}	22.42 ± 1.12^{aC}	26.13 ± 4.68^{aB}
	3.00	27.12 ± 5.72^{aA}	22.14 ± 1.29^{bB}	21.26 ± 0.79^{aB}	24.95 ± 5.27^{aA}
CAT	Control	17.33 ± 1.85^{aA}	16.62 ± 0.59^{aAB}	16.67 ± 0.66^{aAB}	17.33 ± 1.85^{aA}
	0.50	16.50 ± 1.60^{aA}	16.36 ± 1.81^{aA}	15.91 ± 1.76^{aA}	16.50 ± 1.60^{aA}
	1.50	17.66 ± 1.42^{aA}	16.57 ± 1.11^{aA}	16.18 ± 1.01^{aB}	17.66 ± 1.42^{aA}
	3.00	17.17 ± 0.57^{aA}	16.72 ± 1.53^{aA}	16.26 ± 1.49^{aB}	17.17 ± 0.57^{aA}
LPO	control	3.75 ± 0.41^{bAB}	4.31 ± 0.68^{aA}	4.54 ± 0.73^{bA}	2.93 ± 0.32^{bB}
	0.50	3.82 ± 0.21^{abC}	4.83 ± 0.48^{aB}	6.57 ± 0.65^{aA}	2.99 ± 0.16^{abD}
	1.50	3.97 ± 0.26^{abC}	4.96 ± 0.60^{aB}	6.75 ± 0.82^{aA}	3.11 ± 0.21^{abD}
	3.00	4.17 ± 0.33^{aC}	4.80 ± 0.80^{aB}	6.54 ± 1.09^{aA}	3.26 ± 0.26^{aC}

SOD, superoxide dismutase; *GPx*, glutathione peroxidase; *CAT*, catalase; *LPO*, lipid peroxidation. Data ($n = 9$) are presented as mean ± SD. Values with different lowercase letters superscript between control and different concentrations differ significantly ($p < 0.05$) within the same exposure period; values with different uppercase letters superscript differ significantly ($p < 0.05$) among different exposure periods within the same concentration

(A) showed normal hepatic histo-architecture of the piscine liver. The hepatic parenchyma is contained within a thin capsule of fibrous connective tissue. The liver of 0.50% PVC-treated fish showed mild hepatic glycogen depletion while moderate hepatocytic glycogen depletion were observed in 1.50% (C) and 3.0% (D) treated fish. In addition, the liver of 3.0% PVC-treated fish showed marked coagulative hepatocellular necrosis (arrow).

Discussion

The average PVC particle size obtained from the present study was 95.41 ± 4.23 μm (Plate 1) and was observed to be within the environmental relevant size ranges (< 5 mm) found to be ingested by aquatic biota in the wild (Plastics Team 2016; Bråte et al. 2016). Different sizes of PVC microparticles ranging from 20.0 to 250.0 μm have been reported to be ingested

Table 4 Two-way ANOVA statistics for antioxidant enzymes and lipid peroxidation in the liver of *Clarias gariepinus* exposed to poly vinyl chloride MP spiked diets

Source	GPX		SOD		CAT		LPO	
	F	*P* value	*F*	*P* value	*F*	*P* value	*F*	*P* value
Liver								
Corrected model	49.732	0.000	11.242	0.000	2.250	0.027	42.859	0.000
Intercept	28,598.689	0.000	44,026.631	0.000	27,510.122	0.000	6197.830	0.000
Treatment	13.356	0.000*	14.340	*0.062*	.446	0.722	2.937	0.048*
Duration	222.793	0.000*	27.775	0.000*	4.012	0.016*	208.584	0.000*
Treatment * duration	4.171	0.001*	4.698	0.001*	2.263	0.043*	0.925	0.517
Adjusted R squared	0.940		0.766		0.285		0.930	

Values marked with asterisk (*) denotes significant difference ($p > 0.05$). *GPx*, glutathione peroxidase; *SOD*, superoxide dismutase; *CAT*, catalase; *LPO*, lipid peroxidation

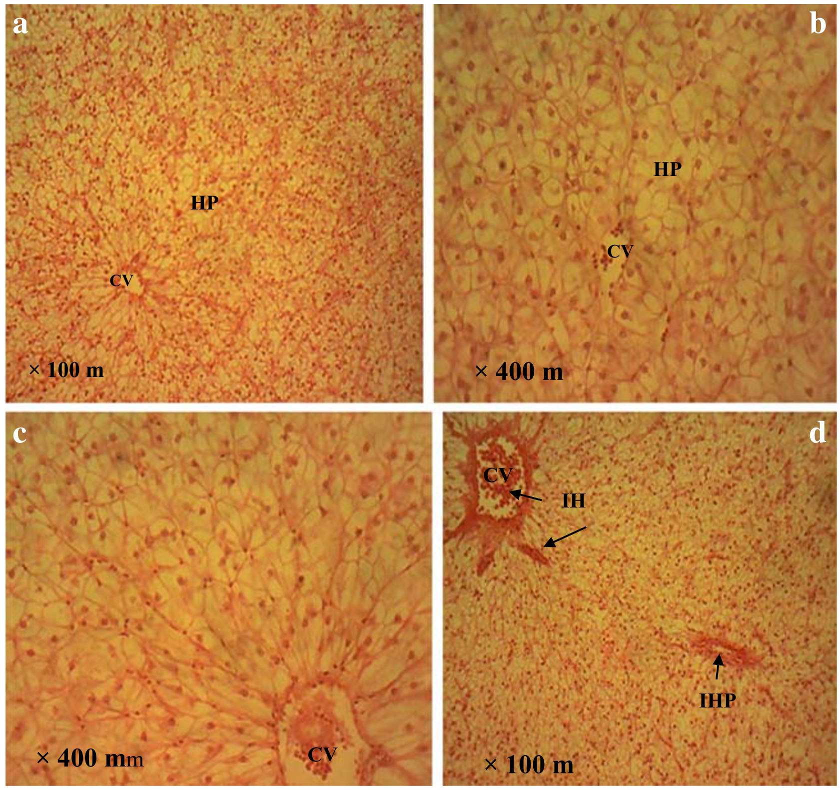

Fig. 2 Liver histopathology of juvenile *Clarias gariepinus* fed different concentrations (%) of PVC MP spiked diets for 15 days. The photomicrographs section A represents control; B, 0.50%; C, 1.50%; D, 3.0% treated group. CV, central vein; HP, hepatic parenchyma; intrahepatic pancreatic tissue (arrows); IH, infiltration of hepatocytes

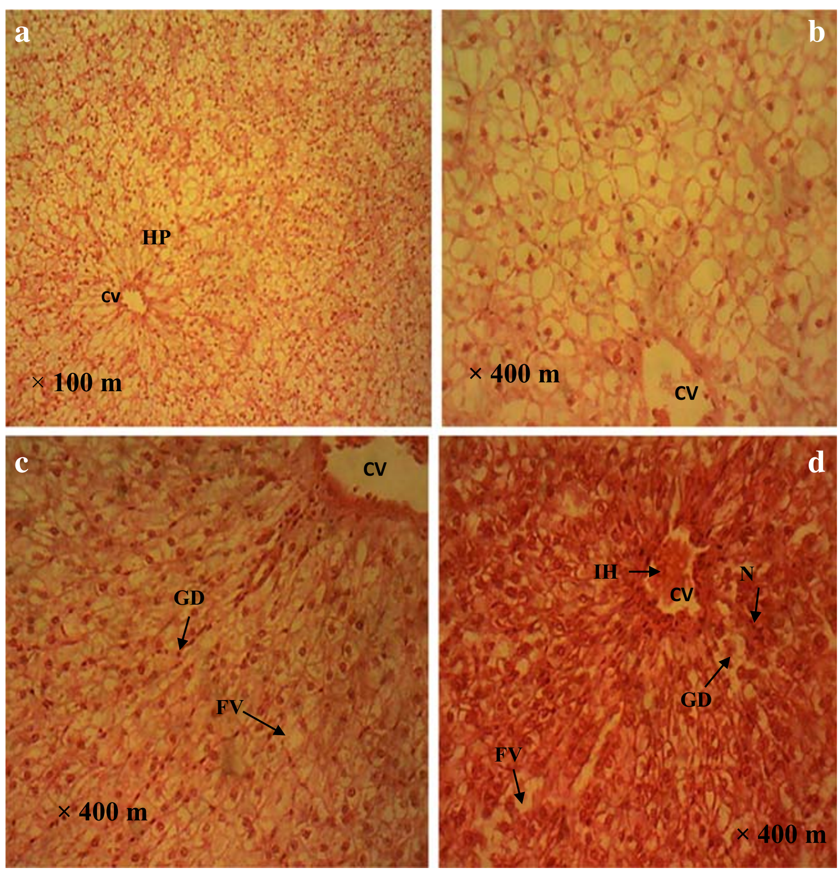

Fig. 3 Liver histopathology of juvenile *Clarias gariepinus* fed different concentrations (%) of PVC MP spiked diets for 30 days. The photomicrographs section A represents control; B, 0.50% PVC-treated fish; C, 1.50%; D, 3.0% treated fish. HP, hepatic parenchyma; CV, central vein; FV, fatty vacuolation; GD, glycogen depletion; N, hepatocellular necrosis; IH, infiltration of hepatocytes

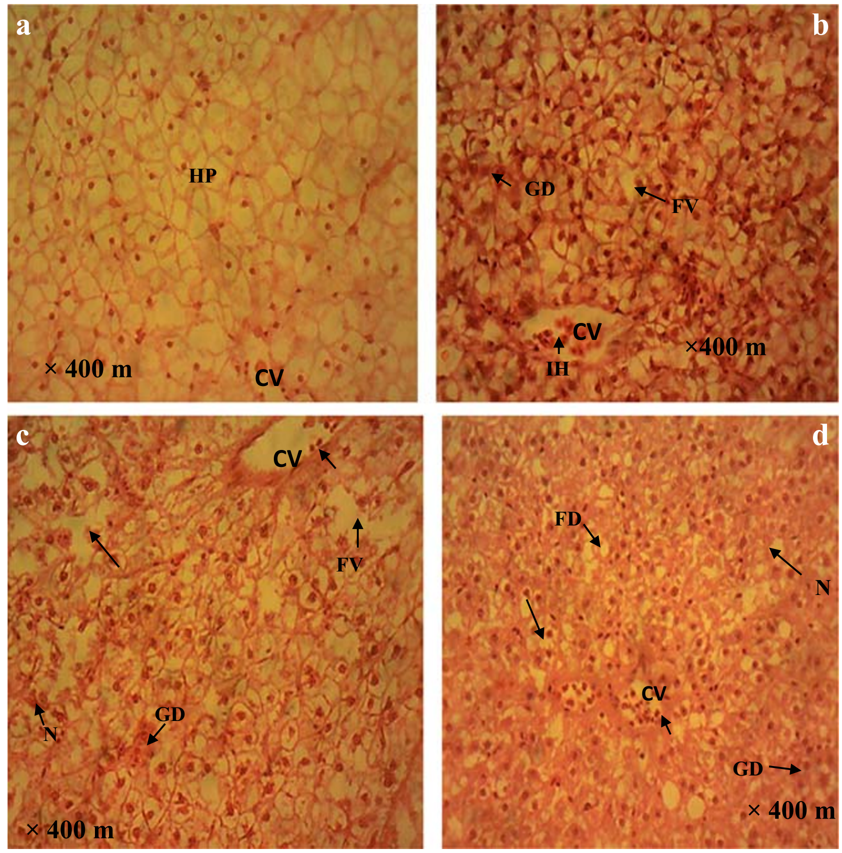

Fig. 4 Liver histopathology of juvenile *Clarias gariepinus* fed different concentrations (%) of PVC MP spiked diets for 45 days. The photomicrographs section A represents control; B, 0.50% PVC-treated fish, C, 1.50%; D, 3.0% treated fish. V, central vein; N, hepatocellur necrosis; FV, fatty vacuolation; FD, fatty degeneration; GD, glycogen depletion

by fish (Jovanovic et al. 2018; Espinosa et al. 2018; Gomiero et al. 2018, 2019).

Organosomatic indices is usually expressed as the apparent ratios of organs to body mass, when the measured organ in relation to body mass can be linked directly to some to changes in the environment (Ronlad and Bruce 1990). According to Dekic et al. (2016), organosomatic indices are expressed through changes in the size of the organ which might be influenced by environmental stressors. The HSI values of PVC-treated groups increased significantly more than the control and may suggest an increased production of hepatocytes and hyperplasia due to the toxic effect of PVC MP. Our current findings deviated from the report of Rummel (2014) who observed marked reduction of HIS in plastic ingested Cod fish (*Gadus morhua*). Critchell and Hoogenboom (2018) reported significant decrease in HSI of Planktivorous fish (*Acanthochromis polyacanthus)* exposed to polyethylene terephthalate microplastic. The variance regarding HSI parameter could possibly stem from the fact that different plastic polymers exert different toxicity action. Assessment of body biomass of fish treated to PVC MP spiked diets after the 45th day showed that PVC had no imminent effect on the specified growth parameter (i.e. fish body weight). In addition, the survival rate after the 45th day exposure showed that the mortality rate was very minimal. Cedervall et al. (2012) reported that the ingestion of polystyrene MP did not affect the growth performance of fish.

The biochemical parameters evaluated in the present study include serum glucose, total protein, triglycerides and serum enzymes (ALT, AST and ALP). Generally, glucose is an important biomarker which is associated with secondary response that help animals survive and recover from challenges such as stress (Barton et al. 2002; Iheanacho et al. 2019). Li et al. (2011) stated that the impact of chemicals on biochemical parameters of fish can provide baseline information about the mechanism and mode of action of these chemicals. Increase in glucose level of PVC-treated groups between the 15th and 30th day exposure suggested an increased utilization of glucose to meet up with the metabolic energy demands which tantamount to stress, properly caused by the PVC toxicant (Saravanan et al. 2012). According to Min and Kane (2008), stress condition is brought by an increase in the corticosteroids which induces hyperglycaemia. The observed reduction in total protein levels in treated groups (0.5% and 3.0%) on the 15th and 30th day exposure could be attributed to an increased utilization of protein to adjust to the escalated metabolic rate which in turn elicited stress and hepatic damage in fish (Saravanan et al. 2012). Reduction in total protein levels are indicative of hepatic damage caused by toxicant-induced stress (Lavanya et al. 2011). Triglyceride levels are useful indicator of liver disease and as

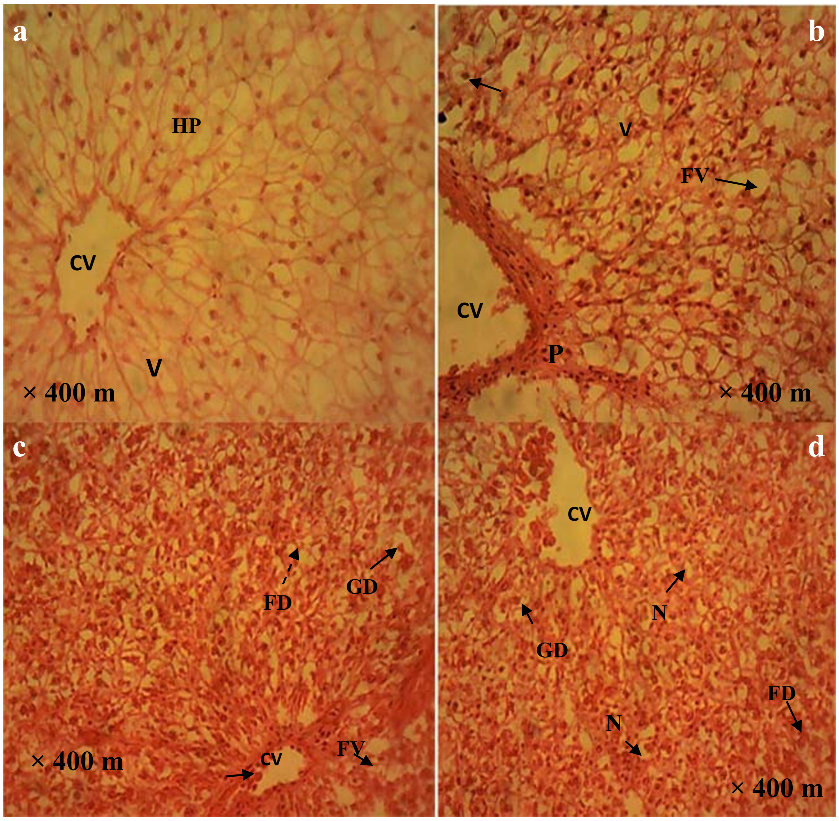

Fig. 5 Liver histopathology in juvenile *Clarias gariepinus* sampled after 30 days of depuration period. The photomicrographs section A represents control; B, 0.50% PVC-treated fish; C, 1.50%; D, 3.0% treated fish. V, central vein; P, portal area, N, necrosis; FD, fatty degeneration; FV, fatty vacuolation; GD, glycogen depletion

well considered to be the foremost index of the health status of teleosts (Wagner and Congleton 2004). Elevation in TG levels of the treated fish groups suggested possible liver disorder caused by PVC toxicant. Subtle biochemical disturbances such as triglyceride fluctuations was observed in dietary titanium dioxide treated fish (*Oncorhynchus mykiss*) (Ramsden et al. 2009). Cedervall et al. (2012) stated that uptake of polystyrene MP by fish altered the ratio of TG to cholesterol in the blood serum level which in turn resulted to fluctuations in the fish metabolism. ALT and AST are mostly used to detect liver disorder (Jung et al. 2003). Elevations in AST and ALT levels of treated groups suggested hepatic damage exerted by PVC toxicant. Epinosa et al. (2017) reported an increase in AST, globulin and albumin levels in *Sparus aurata* fed 0.5 g^{-1} kg^{-1} (0.5%) PVC for 30 days. Jovanovic et al. (2018) reported insignificant changes in AST, ALT, LDH and GGT activities of *S. aurata* fed with 0.1 g kg^{-1} PVC spiked diet. The persistent elevations in AST and ALT levels in PVC-treated fish during the depuration period could be attributed to the prolonged exposure to the toxicant. The decrease in ALT activity of the control on the 15th day exposure could have been caused by subtle stress during fish handling/sampling (Bouck and Ball 1966; Shahsavani et al. 2010).

Antioxidant enzymes are insightful biomarkers of oxidative stress (Egea et al. 2017). Antioxidant enzymes detoxify and annihilate reactive oxygen species (ROS) and other pro-oxidants from the cells under normal circumstance (Ajima et al. 2017). When there is imbalance between the production and elimination of ROS, it brings about oxidative stress especially if the production of ROS exceeds the antioxidant system (Ajima et al. 2017). Song et al. (2006) stated that the adverse effect of oxidative stress involves the alteration in the functions of the antioxidant system, resulting in cell and tissue damage. SOD activity sampled at different exposure periods showed a biphasic trend, suggesting the need to balance the surpassed level of superoxide into a less damaging hydrogen peroxide (Iheanacho and Odo 2020). Lu et al. (2016) stated that ingested polystyrene induced oxidative stress by altering SOD and CAT levels in the gill and liver of Zebra fish *Danio rerio.* The time-dependent inhibition of GPx activity in the liver tissue could be attributed to an increased generation levels of hydroperoxides (Ajima et al. 2017), suspected to have been caused by the adverse effect of PVC MP toxicant. Inhibition of GPx activity in the liver has been reported in *M. galloprovincialis* exposed to polyethylene and polystyrene microplastics (Avio et al. 2015). Ribeiro et al. (2017) reported

Table 5 Histopathological conditions observed in the liver of *Clarias gariepinus* exposed to microplastic polyvinyl chloride MP spiked diet

Exposure period (days)	Concentration (%)	Glycogen depletion	Fatty vacuolation	Hepatocellular necrosis	Infiltration of hepatic cell in central vein	Fatty degeneration
	0.0	–	–	–	–	–
15	0.5	–	–	–	–	–
	1.5	–	–	–	+	–
	3.0	–	–	–	++	–
	0.0	–	–	–	–	–
30	0.50	–	–	–	–	–
	1.5	+	+	–	–	–
	3.0	++	++	+	+++	++
	0.0	–	–	–	–	–
45	0.5	++	++	–	++	++
	1.5	++	++	++	+	++
	3.0	+++	+++	++	++	+++
30 (depuration)	0.0	–	–	–	–	–
	0.5	+	++	–	–	++
	1.5	++	++	+	+	+
	3.0	++	++	+	+	++

–None
+Mild
++Moderate
+++Severe

GPx inhibition in *S. plana* exposed to polystyrene MP. The activity of GPx returned to the normal physiological level in the liver during the depuration phase. The present study revealed that PVC MP orchestrated the inhibition of CAT activities in the liver of 1.5 and 3% exposed fish in a duration-dependent manner. The findings of the present study validated the report of Browne et al. (2013) that PVC MP induced oxyradical production and inhibited CAT activities that resulted in oxidative stress in *Arenicola marina*. Elevations of LPO levels in the liver tissues of PVC-treated groups suggested oxidative damage caused by PVC toxicant. Elevation in LPO level was observed in annelid worm exposed to PVC microparticles (Gomiero et al. 2018).

Histopathology is a biomarker which is useful in studying distinct biological endpoint of historical exposure to challenges (Leonardi et al. 2009). In the present study, no pathological conditions were observed in the liver of the treated groups on the 15th day exposure period (Fig. 2), thus implied that PVC spiked diets caused no imminent harm on the liver during early exposure (Jovanovic et al. 2018). However, mild to moderate glycogen depletion and fatty vacuolation observed in 1.50% and 3.0% PVC-treated groups on the 30th day exposure (Fig. 3) could be attributed to the interference PVC toxicant on carbohydrate metabolism and is linked towards the amount of energy needed for detoxification (Hugla and Thome 1999; Rochman et al. 2013). Further exposure (45 days) was marked with severe glycogen depletion (Fig. 4B–D), fatty liver generation (Fig. 4B–D) and moderate hepatocellular necrosis (Fig. 4D). These biological responses indicated that the liver of the treated groups was damaged due to the prolonged exposure to PVC toxicant. Espinosa et al. (2019) reported similar histopathological changes such as increase in hepatocyte vacuolation, loss of organisation of parenchyma, congestion of blood sinusoids and focal necrosis in the liver of PVC-MP fed fish (*Dicentrarchus labrax*). Lu et al. (2016) reported histopathological changes such as lipid accumulation and marked inflammation in the liver of Zebrafish (*Danio rerio*) exposed to microplastic. Jovanovic and Palic (2012) reported an inflammation and oxidative stress in fish exposed to nanoplastics. Rochman et al. (2013) reported histopathological changes such as glycogen depletion, single-cell necrosis, fatty vacuolation and fatty degeneration in the liver of Japanese medaka (*Oryzia slatipes*) treated with MP spiked diet. The marked alterations seen in the liver of the treated groups (Fig. 5B–D), even at the end of the depuration period suggested that the fish are yet to fully recover from the hepatic damage caused by prolonged exposure to PVC MP.

Conclusion

The findings of the present study revealed that dietary exposure to polyvinyl chloride MP undermined the health of the exposed fish. A severe liver damage was observed in fish exposed to PVC MP after the careful analysis of histological alterations and evident modifications of the activity of serum enzymes.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

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